S1 Supplementary Methods

Measurement contract, occupancy scalar, matching rule, LOR labels, and statistical design

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S1.0 Dataset and measurement contract

Cambridge propofol sedation resting-state EEG, 20 healthy adults, four protocol phases each (Chennu et al., 2015; 2016). Source data DOI 10.17863/cam.68959. This article consumes a frozen NeuralManifoldDynamics HDF5 export, `neuralmanifolddynamics_Sedation-RestingState_20260918_075713` (80 files; pinned example `sub-002_sedation_rest_run-003_acq-135.h5`, SHA-256 `1635513f690ae01807a9495d920052ddc2a2c9a568d3b8a76bb04139bca993ba`). Article `datainfo.mat` pin: SHA-256 `ea8824f3ca9bd2eef0cdee3a7e1402dcc9e599f9688290cc382ae8e1a21f5608`. The article-local `reproducibility_manifest.json` records repository base commit `d1c7fcedfb92c34162437fe11cdb35aa64620018`, result-manifest hashes, and the limitation that the collection manifest does not hash every raw H5 file; the current revision remains uncommitted.

Analyses should be read against the export's measurement contract, not as direct computations on raw voltage traces. Exported `coords_3d_subject_anchored` uses local fit populations with ComBat disabled; confirmatory amplitude claims on that chart remain gated. Occupancy does not use that coordinate. It uses eight MNPS-used `features_raw` columns in a baseline-only robust-z frame constructed article-side.

Protocol phases come from `datainfo.mat` column 2 and are not inferred from `run`/`acq` order or from H5 `condition=sedation`. Column 3 is plasma concentration (μg/L), column 4 mean reaction time (ms), column 5 k correct / 40. The package alias `unresponsive` for protocol order 3 is retained only as `datainfo_legacy_condition` and is not an LOR label.

S1.1 Epoch structure and features

Feature windows are nominally 8 s with a 4 s step (`observed_dt` typically 4 s; occupancy increments require $\Delta t \in [3.8, 4.2]$ s). Occupancy features:

`eeg_delta`, `eeg_theta`, `eeg_alpha`, `eeg_beta`, `eeg_gamma`, `eeg_alpha_theta`, `eeg_beta_alpha`, `eeg_sample_entropy`.

These are the eight spectral/complexity columns used by the MNPS feature contract in this export. Occupancy is median $\|z\|_2$ after baseline-only robust standardisation, not a three-dimensional MNPS radius.

The article-side reanalysis consumes these frozen feature values and does not re-run EEG preprocessing or estimate PSDs. Channel handling, reference, artefact treatment, band edges, PSD estimator, sample-entropy parameters, and any gamma-EMG safeguards are therefore properties of the pinned export, not newly estimated methods in this article. They must be audited from the export-producing pipeline before any claim beyond this frozen feature contract; in particular, the gamma contribution reported here is descriptive rather than an EMG-robust physiological claim.

S1.2 Occupancy and matching

For participant i, all baseline rows with finite values in all eight features define

$$m_j = \mathrm{median}\left(x_{\cdot,j}^{\mathrm{base}}\right), \quad s_j = 1.4826\,\mathrm{MAD}\left(x_{\cdot,j}^{\mathrm{base}}\right), \tag{1}$$

Scales that are non-finite or $\leq 10^{-8}$ are set to invalid (`NaN`); no floor value is substituted or clamped. This scaling step is separate from one-step eligibility. Later-phase rows are mapped $z_{t,j} = (x_{t,j} - m_j)/s_j$. Occupancy is the median Euclidean radius of eligible one-step sources (finite features and valid Δt). This is the unmatched occupancy scalar copied from the baseline-anchor matching pipeline; it is not recomputed under a new distance.

Matching is greedy one-to-one Euclidean in z, caliper = median baseline leave-one-out nearest-neighbour distance, minimum 20 pairs per participant, minimum 10 included participants for a cohort test. Increment speed

and spread are summaries of $\Delta z/\Delta t$ on matched or unmatched eligible sources (S2). Matching is outcome-blind.

Recording-local `features_robust_z` is centred at 0 in every recording, so occupancy is invisible in that chart. Local-chart MNPS matching therefore cannot be read as a cross-phase physical-state match.

S1.3 LOR/ROR

Exact binomial 95% Clopper–Pearson intervals (Clopper & Pearson, 1934) on k out of 40 at baseline versus later protocol sedation. LOR-impaired if the later interval lies entirely below baseline; otherwise responsive (Chennu et al., 2016). Among impaired participants, ROR returned if the recovery interval overlaps baseline. Labels are assigned from k only, then joined to frozen occupancy.

S1.4 Statistical design

Unit of analysis: participant. The initially selected first-sedation-minus-baseline contrast is an ordered reporting contrast, not a confirmatory primary endpoint. Its exact two-sided sign test on non-zero differences, Wilcoxon signed-rank p, paired d_z, and percentile bootstrap 95% CI (10 000 participant resamples, seed 20260920) are descriptive. Later sedation and recovery are ordered descriptive contrasts. LOR group tests use two-sided Mann–Whitney and two-sample d, with a gate ≥ 10 per arm. Association: Spearman ρ occupancy versus k at later sedation. Discovery local-chart screens: Holm on original H1/H2 sign tests; Benjamini–Hochberg FDR on a frozen Wilcoxon family (S2). No FDR survivor is promoted.

S1.5 Exploratory status and gates

Dated preregistration draft 2026-09-20 specified G0–G4 before confirmatory contrasts. G0 inventory, G1 labels, and G2 support were closed on this export. G3 independent code review and G4 lock were not completed. Occupancy promotion is post-hoc relative to that draft for first-sedation and recovery values. All p-values in the article are descriptive and exploratory.

S1.6 What was not done

No new occupancy definition, no lowering of the $n = 10$ matched-subject gate, no treating protocol 3 as LOR, and no confirmatory H1/H2 amplitude claim on `subject_anchored` MNPS were made. A separate post-hoc robustness branch (`results/v1_occupancy_robustness_20260922_reviewfix/`) did decompose the frozen radius and tested leave-one-feature-out, baseline-only sample-SD, and normalised-IQR scales. It reused the original full-feature eligibility mask, participant unit, and missingness; it therefore quantifies computational sensitivity for these choices but does not establish biological robustness, independence, or confirmatory validity. No consciousness inference.

The robustness branch is computationally locked by its hash manifest, while G3 independent review and G4 confirmatory locking remain open.

Bibliography

Chennu, S., O'Connor, S., Adapa, R., Menon, D. K., & Bekinschtein, T. A. (2015,). *Research data supporting ``Brain connectivity during propofol sedation''*. Apollo – University of Cambridge Repository. https://doi.org/10.17863/cam.68959

Chennu, S., O'Connor, S., Adapa, R., Menon, D. K., & Bekinschtein, T. A. (2016). Brain Connectivity Dissociates Responsiveness from Drug Exposure during Propofol-Induced Transitions of Consciousness. *PLOS Computational Biology*, *12*(1), e1004669. https://doi.org/10.1371/journal.pcbi.1004669

Clopper, C. J., & Pearson, E. S. (1934). The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial. *Biometrika*, *26*(4), 404–413. https://doi.org/10.1093/biomet/26.4.404